

Notes: Busse and Green (JFE, 2002)

Market Efficiency in Real Time

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	CNBC call reports	3
3.2	Trades, quotes, and prices	3
3.3	Trade direction and order imbalance	4
4	Models and empirical specifications	4
4.1	Event-time return measurement	4
4.2	Bootstrap inference for intraday returns	5
4.3	Cross-sectional price-impact regressions	5
4.4	Trading-profit calculation	5
5	Identification and empirical design	5
5.1	What the design identifies	5
5.2	Selection, leakage, and timing concerns	6
5.3	Microstructure concerns	6
6	Main findings	6
6.1	Price response magnitude	6
6.2	Speed of price adjustment	7
6.3	Cross-sectional price impact	7
6.4	Trading activity and order flow	7
6.5	Short-horizon profitability	8
7	Tables and figures	8
7.1	Table 1: Descriptive statistics of CNBC call reports	8
7.2	Figure 1: Stock price reaction to CNBC reports	8
7.3	Table 2: Speed of price adjustment to CNBC reports	9
7.4	Table 3: Characteristics of price impact	9
7.5	Table 4: Daily cumulative abnormal returns following CNBC reports	10
7.6	Figure 2: Trading profits around CNBC reports	10
7.7	Table 5: Trading intensity around CNBC reports	11
7.8	Table 6: Order imbalance around CNBC reports	11
7.9	Table 7: Fraction of NYSE-listed trades executed off the NYSE	12
8	Short summary	13

9	Conclusion	13
9.1	Core conclusion	13
9.2	Economic intuition	14
9.3	Incremental contribution revisited	14
9.4	Implications for market efficiency	14
9.5	Limitations	14
9.6	Takeaway	14

1 Big picture

- **Question.** How fast do security prices incorporate widely disseminated information when that information arrives during the trading day? The paper asks whether market efficiency should be measured in days, minutes, or seconds.
- **Setting.** The authors study analyst opinions reported during CNBC's *Morning Call* and *Midday Call* segments. These reports air while the stock market is open, so the authors can observe second-by-second stock-price and trading responses.
- **Data contribution.** The paper combines hand-collected video timestamps of CNBC stock mentions with intraday TAQ trades and quotes. Time zero is the second when the stock ticker or company name is first shown or spoken.
- **Sample.** The sample contains 322 stock reports over 84 trading days from June 12, 2000 through October 27, 2000. The authors classify each report as positive or negative and exclude mixed-sentiment reports.
- **Main price result.** Positive reports during the *Midday Call* have a large, statistically significant price impact that begins within seconds and is mostly incorporated within one minute.
- **Asymmetric adjustment.** Negative reports also move prices in the expected direction, but the response is slower and continues for roughly 15 minutes. The paper interprets this as potentially related to the higher costs and frictions of short selling.
- **Trading result.** Trading intensity more than doubles immediately after the reports. Order imbalances move in the direction of the news: buyer-initiated trades rise after positive reports, and seller-initiated trades rise after negative reports.
- **Profitability result.** Traders who buy very quickly after positive *Midday Call* reports can earn small but statistically significant profits after accounting for trading costs. The profitable window is extremely short: roughly the first 15 seconds.
- **Economic takeaway.** Markets are very efficient, but not literally instantaneous. Active traders who monitor real-time news help make markets efficient, and the compensation for doing so is small, short-lived trading profit.

2 Incremental contribution of the paper

- **From minutes to seconds.** Earlier market-efficiency studies often conclude that prices adjust within five to 15 minutes. This paper shows that, for liquid stocks and real-time broadcast news, the economically relevant adjustment window can be seconds.
- **Real-time information release.** Most analyst-recommendation studies use daily or event-date data. This paper studies information released while the market is open and records the exact airing time, allowing much sharper inference about speed.
- **Media as an information channel.** The paper treats CNBC as a fast public dissemination mechanism for analyst opinions. It therefore contributes to understanding how financial media transmit information into prices.

- **Integration of price, volume, and order flow.** The paper does not only measure returns. It also studies trading intensity, buyer/seller order imbalance, and whether traders can profit after accounting for the bid–ask spread.
- **Evidence on active traders.** The results are consistent with the Grossman–Stiglitz idea that informed monitoring is costly and that small positive profits compensate traders who process information immediately.
- **Positive-versus-negative asymmetry.** The paper documents that positive news is incorporated quickly, while negative news is slower. This adds evidence on how short-sale frictions and execution constraints shape intraday price discovery.
- **Cross-sectional evidence on information content.** The paper studies which reports have larger price impact. *Midday Call* reports, reports with price charts, technology/internet firms, firms with negative EPS, and smaller firms show larger effects.
- **Trading-feasibility contribution.** By examining off-NYSE execution for NYSE-listed stocks, the paper connects the profitability evidence to how traders try to obtain fast execution.

3 Data and measurement

3.1 CNBC call reports

- The paper studies CNBC’s *Morning Call* and *Midday Call* analyst-call segments.
- The *Morning Call* usually airs around 11:05–11:10 a.m. Eastern time; the *Midday Call* usually airs around 2:53–2:58 p.m. Eastern time.
- The sample period covers June 12 through October 27, 2000.
- The authors videotape CNBC and record the exact initial mention time of each stock, using TV Guide Channel time stamps and `time.gov` to calibrate the clock.
- Time zero is the first second when the ticker symbol is shown or spoken; if no ticker is given, it is the first second when the company name is disclosed.
- The authors record whether the report is positive or negative, whether a price chart is displayed, and contemporaneous market-index conditions.

Interpretation: The central measurement advantage is timing. Because the reports occur during market hours and are timestamped to within a few seconds, the paper can examine adjustment at horizons that daily event studies cannot see.

3.2 Trades, quotes, and prices

- Intraday trades and quotes come from the NYSE TAQ database.
- The main intraday return measure is the percentage change in the prevailing midquote, defined as the average of the bid and ask prices.
- The authors use primary-market quotes to avoid passive autoquotes that may overstate available liquidity or trading costs.

- For daily abnormal returns, the paper uses two benchmarks: a size-and-SIC matched portfolio and a market-model residual based on the CRSP value-weighted NYSE/AMEX/Nasdaq market portfolio.

Interpretation: Using midquotes reduces bid–ask bounce in the return measurement. Using TAQ trades and quotes allows the paper to evaluate both the speed of price adjustment and the trading costs faced by real-time traders.

3.3 Trade direction and order imbalance

Trades are classified as buyer- or seller-initiated by comparing transaction prices to prevailing quotes and using the tick rule for trades inside the bid–ask spread. The paper measures order imbalance as

$$IMBAL_{it} = \frac{nBuys_{it} - nSells_{it}}{nBuys_{it} + nSells_{it}}, \quad (1)$$

where $nBuys_{it}$ and $nSells_{it}$ are the numbers of buyer- and seller-initiated trades for report i during interval t .

- $IMBAL = 1$ means all trades are buyer-initiated.
- $IMBAL = -1$ means all trades are seller-initiated.
- A positive report should generate positive order imbalance if viewers are buying on the news.
- A negative report should generate negative order imbalance if viewers are selling on the news.

Interpretation: Order imbalance is the paper’s bridge from price response to trading behavior. It shows whether the price response is accompanied by order flow consistent with viewers trading on the information.

4 Models and empirical specifications

4.1 Event-time return measurement

Let $P_{i,t}$ denote the midquote for stock i at event time t . An intraday return over an interval $[a, b]$ is measured as

$$R_{i,[a,b]} = \frac{P_{i,b} - P_{i,a}}{P_{i,a}}. \quad (2)$$

The figures and tables cumulate average returns around the CNBC report time.

- For positive reports, the key adjustment window is the first minute after the initial mention.
- For negative reports, the key adjustment window is much longer, roughly 15 minutes after the initial mention.
- The paper reports separate patterns for positive/negative reports and *Morning Call/Midday Call* reports.

Interpretation: The paper’s event time is unusually precise. This allows it to distinguish an adjustment that occurs within 15 seconds from one that occurs within one minute or 15 minutes.

4.2 Bootstrap inference for intraday returns

Because intraday returns are not normally distributed, the paper uses a nonparametric bootstrap to compute empirical p -values.

- The empirical distribution puts probability $1/n$ on each observed cross-sectional return.
- The authors draw 10,000 bootstrap samples with replacement.
- The bootstrap distribution of the mean return is used to test whether the average return differs from a benchmark such as zero.

Interpretation: Bootstrap inference is important because the return intervals are very short and intraday returns can be skewed, heavy-tailed, and influenced by microstructure noise.

4.3 Cross-sectional price-impact regressions

The paper estimates cross-sectional regressions of the form

$$Return_i = \alpha + \beta_1 MIDDAY_i + \beta_2 CHART_i + \beta_3 NUMREPORTS_i + \Gamma' FirmChar_i + \varepsilon_i. \quad (3)$$

- For positive reports, the dependent variable is the one-minute return after the report.
- For negative reports, the dependent variable is the 15-minute return after the report.
- *MIDDAY* indicates the *Midday Call*; *CHART* indicates that a historical price chart is displayed.
- Firm characteristics include technology, internet, negative EPS, and size-quintile indicators.

Interpretation: These regressions ask which reports are more informative. A larger coefficient means that the market reacts more strongly to that type of report.

4.4 Trading-profit calculation

For positive *Midday Call* reports, the paper computes returns from buying at different intervals around the initial mention and selling between one and two minutes after the mention.

- Buy prices are based on buyer-initiated transaction prices.
- Sell prices are based on seller-initiated transaction prices.
- This construction accounts for the bid–ask spread and therefore approximates executable trading profits.

Interpretation: The profitability exercise is more demanding than a raw-return calculation. It asks whether a trader could have captured the price response after paying realistic execution costs.

5 Identification and empirical design

5.1 What the design identifies

- The paper identifies the very short-run price and trading response to a widely disseminated information event.

- The design is strongest for measuring speed: exact initial mention times allow the authors to evaluate whether prices move in seconds or minutes.
- The event window approach compares the same stock immediately before and after the broadcast, limiting the role of slow-moving firm characteristics.
- The combination of price, volume, order imbalance, and trading-profit evidence supports the interpretation that viewers actively trade on the reports.

Interpretation: The paper is not just saying that CNBC reports are associated with price moves. It shows that price moves and order-flow shifts occur immediately after the information is broadcast.

5.2 Selection, leakage, and timing concerns

- The CNBC reports are not randomly assigned. Analysts and anchors select which stocks to discuss, and stocks may already be receiving attention.
- There is evidence of pre-report run-up before positive *Midday Call* reports, consistent with some market participants learning the information before the CNBC airing.
- The sentiment classification is subjective, although the authors report that the tone is usually unambiguous and mixed reports are excluded.
- The negative-report sample is small, so estimates for negative news have lower statistical power.
- The paper focuses on large, liquid firms. The estimated speed of adjustment may be an upper bound on how fast prices incorporate news for less liquid securities.

Interpretation: The key causal object is not the full value of analyst information from first private discovery. Rather, it is the incremental market response when that information becomes broadly and visibly disseminated through CNBC.

5.3 Microstructure concerns

- Very short horizons make trade/quote synchronization important.
- The authors test alternative trade-classification algorithms, including Lee–Ready-style lagged quote comparisons and other timing adjustments.
- They also examine off-NYSE execution for NYSE-listed stocks, because fast traders may route orders to venues that execute quickly at the National Best Bid or Offer.

Interpretation: The paper’s microstructure checks matter because the main profitability result depends on execution within seconds. Small timestamp and quote-matching errors could otherwise change the inference.

6 Main findings

6.1 Price response magnitude

- Positive *Midday Call* reports generate a large price increase. The cumulative return from 15 minutes before to one minute after the mention is about 62 basis points, with 41 basis points occurring during the first minute after the mention.

- Seventy-three percent of firms discussed positively during the *Midday Call* have positive one-minute returns, with an average market-capitalization increase of about \$109 million.
- Positive *Morning Call* reports have a much smaller response, suggesting that those reports are less new, less relevant, or already reflected in prices.
- Negative reports have larger but slower price responses. The 15-minute returns are about -93 basis points for *Morning Call* negative reports and -75 basis points for *Midday Call* negative reports.

Interpretation: The price response is economically meaningful. CNBC is not merely entertainment or noise in this setting; at least some reports contain information that moves prices.

6.2 Speed of price adjustment

- Prices react significantly in the first 15 seconds after positive and negative reports.
- For positive reports, the response continues for roughly 45–60 seconds and then largely stops.
- For negative reports, price adjustment continues gradually for roughly 15 minutes.
- The faster positive-news response is consistent with easier implementation of long positions than short positions.

Interpretation: The paper narrows the practical meaning of market efficiency. For positive liquid-stock news, one minute can be enough for public information to be incorporated.

6.3 Cross-sectional price impact

- *Midday Call* reports have a larger price impact than *Morning Call* reports.
- Reports accompanied by a price chart have a larger impact, consistent with more detailed reports carrying more information.
- Technology and internet firms have larger price responses.
- Firms with negative EPS have larger price responses, consistent with analyst information being especially valuable when fundamentals are hard to evaluate.
- Smaller firms show larger responses, consistent with lower prior information availability.

Interpretation: The market reacts more when the report is likely to contain more information or when the firm is harder to value using public fundamentals.

6.4 Trading activity and order flow

- Trading intensity rises sharply after the reports.
- For positive *Midday Call* reports on Nasdaq stocks, median trades per minute rise from 49 in the five minutes before the report to 286 in the first minute after the report.
- Order imbalances shift in the direction of the report sentiment.
- After positive *Midday Call* reports, median order imbalance in the first minute is 0.50 for NYSE stocks and 0.398 for Nasdaq stocks.

- After negative reports, first-minute order imbalance is negative, about -0.487 for NYSE stocks and -0.330 for Nasdaq stocks.

Interpretation: The trading results show that the price movements are accompanied by aggressive trading in the direction of the news. This is consistent with viewers or other fast traders reacting almost immediately.

6.5 Short-horizon profitability

- Traders who know the information before the broadcast can earn the largest profits. Buying four to five minutes before positive *Midday Call* reports and selling one to two minutes after yields about 60 basis points after trading costs.
- Buying within the first five seconds after the report yields about 40 basis points.
- Buying within the first 15 seconds yields about 16 basis points.
- Beyond about 15 seconds, profits are no longer statistically significant; beyond about 25 seconds, average profits are negative.

Interpretation: The profit window exists, but it is extremely narrow. This is exactly the kind of small, rapidly competed-away profit predicted by costly-information theories of market efficiency.

7 Tables and figures

7.1 Table 1: Descriptive statistics of CNBC call reports

Read:

- The sample has 174 *Morning Call* reports and 148 *Midday Call* reports.
- Positive reports are much more common than negative reports: 155 versus 19 for the *Morning Call*, and 125 versus 23 for the *Midday Call*.
- Price charts are much more common during the *Midday Call*: 78 reports show charts compared with only 11 during the *Morning Call*.
- The firms are large. Median market capitalization is \$47.8 billion for *Morning Call* firms and \$16.7 billion for *Midday Call* firms.

Economic message: The setting is a sample of large, liquid stocks, mostly receiving positive analyst coverage. This makes it a favorable environment for very fast price discovery.

7.2 Figure 1: Stock price reaction to CNBC reports

Read:

- The figure plots cumulative returns from 15 minutes before to 15 minutes after the initial mention.
- Positive *Midday Call* reports show a sharp increase immediately after time zero.
- Positive *Morning Call* reports show little cumulative response.
- Negative reports decline after the mention, with a much more gradual and persistent pattern.

Economic message: The figure visually establishes the paper’s main result: positive afternoon news is capitalized within about a minute, while negative news diffuses more slowly.

Table 1
Descriptive statistics of CNBC call reports
The table shows characteristics of on-air stock reports during the Morning Call and Midday Call segments on CNBC. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

	Morning Call	Midday Call
<i>Panel A: Report characteristics</i>		
Stock reports	174	148
Average reports per segment	4.3	2.3
Positive reports	155	125
Negative reports	19	23
Price chart displayed	11	78
Price chart not displayed	163	70
Nasdaq index up for the day	85	79
Nasdaq index down for the day	89	69
Dow index up for the day	74	89
Dow index down for the day	100	59
<i>Panel B: Firm characteristics</i>		
NYSE-listed	102	109
Nasdaq-listed	71	39
AMEX-listed	1	0
Median market capitalization (in \$millions)	47,807	16,680

Table 1: Descriptive statistics

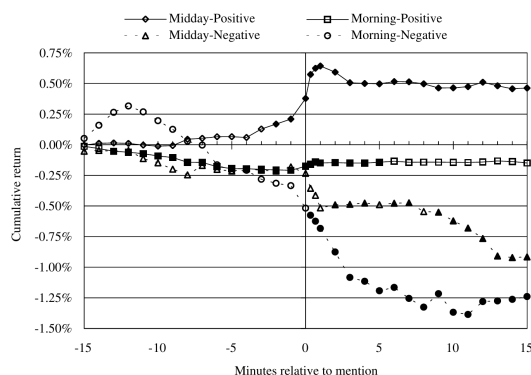


Fig. 1. Stock price reaction to CNBC reports. The figure shows the reaction of stock prices to on-air stock reports during the Morning Call and Midday Call segments on CNBC. The chart plots cumulative returns beginning 15 minutes before the stock report. The plot partitions reports into groups depending on whether the reports are positive or negative. A solid marker denotes that the cumulative average return is significantly different from zero at the five percent level using bootstrap empirical p -values. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

Figure 1: Stock price reaction

7.3 Table 2: Speed of price adjustment to CNBC reports

Read:

- The table reports price changes over very short intervals after the report.
- Positive *Midday Call* reports have significant positive returns in the first 15, 30, 45, and 60 seconds.
- Positive *Morning Call* reports have smaller but sometimes significant short-horizon returns.
- Negative reports show significant negative returns in early intervals and additional declines over longer windows.

Economic message: The table shows that the relevant unit of market adjustment is seconds, not trading days. For positive news, most adjustment is complete inside one minute.

7.4 Table 3: Characteristics of price impact

Read:

- For positive reports, the *Midday Call* dummy is positive and significant, indicating larger price impact in the afternoon segment.
- The chart dummy is positive and significant, suggesting more detailed reports have more information content.
- Technology, internet, negative EPS, and small-firm indicators increase the price impact of positive reports.
- For negative reports, technology, internet, and negative EPS are associated with larger negative effects, but the sample is small.

Economic message: The market response is larger when the report is more likely to be informative and when firm valuation is more uncertain.

Table 2

Speed of price adjustment to CNBC reports

The table reports average percentage price changes following on-air stock reports during the Morning Call and Midday Call segments on CNBC. We calculate price changes over each interval, where the interval begins where the previous interval ends. For example, the interval labeled 0.25 corresponds to the price adjustment from 0 to 0.25 minutes. One and two stars denote significance at the five and one percent level, respectively, using bootstrap empirical p -values. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

Minutes relative to report	Morning Call		Midday Call	
	Positive reports	Negative reports	Positive reports	Negative reports
-15 to -10	-0.073%	0.268%	-0.003%	-0.113%
-10 to -5	-0.106**	-0.431**	0.068	-0.090
-5 to 0	-0.030	-0.170	0.143*	0.020
0.25	0.024**	-0.136**	0.108**	-0.044*
0.50	0.014*	-0.072	0.193**	-0.066*
0.75	0.014	-0.045	0.075**	-0.073**
1.00	0.016*	-0.040	0.037*	-0.051**
1.25	-0.007	0.018	0.017	-0.030
1.50	0.002	0.004	0.015	-0.067*
1.75	-0.011	-0.076*	0.007	-0.005
2.00	0.007	-0.005	-0.021	0.001
2.25	0.010	-0.050	-0.021	0.012
2.50	-0.005	-0.025	-0.014	0.006
2.75	-0.001	-0.037*	-0.019	0.003
3.00	-0.002	-0.084*	0.004	0.006
3.25	0.005	-0.040	-0.023	-0.014
3.50	-0.004	-0.065*	-0.021	0.019
3.75	-0.008	-0.025	-0.024	0.014
4.00	0.005	-0.080**	-0.016	-0.016
4.25	-0.006	-0.008	-0.010	-0.002
4.50	-0.006	0.024	0.001	-0.001
4.75	-0.001	0.003	-0.015	-0.009
5.00	0.012	-0.048*	0.018*	0.023*
10.00	0.007	-0.102	-0.036	-0.076
15.00	0.005	-0.047	-0.007	-0.380**
30.00	0.035	-0.067	0.027	0.146

Table 2: Speed of price adjustment

Table 3

Characteristics of price impact of CNBC reports

The table shows results of cross-sectional regressions of percentage returns following on-air stock reports during the Morning Call and Midday Call segments on CNBC. Panel A reports the results for positive reports and uses one-minute returns. Panel B reports the results for negative reports and uses 15-minute returns. MIDDAY is 1 if the stock report is during the Midday Call segment and 0 if the report is during the Morning Call. CHART is 1 if a historical stock price chart is displayed during the report and 0 otherwise. NUMBER OF REPORTS is the number of stocks discussed during that day's segment. TECHNOLOGY and INTERNET are industry dummy variables classified using the North American Industry Classification System, and are 1 if the firm operates in that industry, 0 otherwise. NEGATIVE EPS is 1 if the stock has a negative EPS at the time of the report, 0 otherwise. We construct the SIZE QUINTILE dummy variables from CRSP NYSE market capitalization quintiles, and they are 1 if the market capitalization of the firm falls into the designated quintile, 0 otherwise. We calculate standard errors using White's correction for heteroskedasticity. One and two stars denote significance at the five and one percent level, respectively. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

<i>Panel A: Positive reports</i>			
Observations	280	276	276
Adjusted R^2	0.160	0.248	0.405
Coefficients			
CONSTANT	0.146*	-0.047	-0.026
MIDDAY	0.202**		0.234**
CHART	0.236**		0.198**
NUMBER OF REPORTS	-0.017*		-0.016
TECHNOLOGY		0.178**	0.171**
INTERNET		0.439*	0.439*
NEGATIVE EPS		0.532*	0.550**
SIZE QUINTILE 1-3		0.827**	0.857**
SIZE QUINTILE 4		0.337*	0.226
<i>Panel B: Negative reports</i>			
Observations	42	42	42
Adjusted R^2	0.054	0.182	0.172
Coefficients			
CONSTANT	-0.889**	-0.334*	0.037
MIDDAY	0.284		-0.350
CHART	-0.719		-0.371
NUMBER OF REPORTS	-0.022		0.006
TECHNOLOGY		-0.855**	-0.970*
INTERNET		-1.537**	-1.197**
NEGATIVE EPS		-1.068**	-1.168**
SIZE QUINTILE 1-3		0.552	0.462
SIZE QUINTILE 4		0.000	-0.212

Table 3: Characteristics of price impact

7.5 Table 4: Daily cumulative abnormal returns following CNBC reports

Read:

- The table compares longer-horizon abnormal returns using both size/SIC matching and a market-model benchmark.
- Positive *Midday Call* reports have positive abnormal returns on the day of the report and shortly after, especially using the size/SIC benchmark.
- Negative *Midday Call* reports have increasingly negative abnormal returns over longer horizons.
- *Morning Call* evidence is weaker and less robust.

Economic message: The intraday response is not obviously just temporary noise. Some negative reports appear to be the beginning of a longer deterioration in firm value.

7.6 Figure 2: Trading profits around CNBC reports

Read:

- The figure plots trading profits from buying at different times and selling one to two minutes after positive *Midday Call* reports.
- Profits are highest for traders who buy before the report airs.

- Profits remain positive for traders who execute almost immediately after the first mention.
- The profit surface slopes down rapidly after time zero.

Economic message: Fast execution is essential. The figure turns the idea of market efficiency into a practical trading statement: the opportunity disappears within seconds.

Table 4
Daily cumulative abnormal returns following CNBC reports
The table reports daily cumulative abnormal percentage returns following on-air stock reports during the Morning Call and Midday Call segments on CNBC. We calculate cumulative returns by comparing the midquote price 15 minutes before the report to the closing prices on the days following the report. In Panel A, we take as the abnormal return the difference between the total stock return and the return on an equal-weighted portfolio of stocks with the same SIC code and from the same NYSE market capitalization decile. In Panel B, we use the residual from a market model regression as the abnormal return. We base the cumulative abnormal return standard errors on stock residuals over a 250-day estimation period prior to the day of the report. One and two stars denote significance at the five and one percent level, respectively. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

Days relative to report	Morning Call		Midday Call	
	Positive reports	Negative reports	Positive reports	Negative reports
<i>Panel A: Size, SIC</i>				
0	0.280%	-2.550%	1.174%**	-0.529%
1	0.541	1.568	1.402*	-1.333
2	0.726	1.200	1.254	-2.219
3	1.530*	1.141	1.574	-3.503
4	1.497	1.971	1.655	-2.849
5	1.465	0.506	1.120	-2.520
6	0.984	2.328	1.509	-4.101
7	0.635	0.371	1.410	-4.718
8	0.627	-0.740	1.676	-5.072
9	0.277	-0.425	1.445	-7.435**
10	0.226	-1.594	1.391	-7.535*
15	-0.872	-5.585	2.241	-11.519**
20	-1.113	-6.471	0.408	-11.519**
<i>Panel B: Market model</i>				
0	0.293%	-0.618%	0.794%*	-0.516%
1	0.648	2.708	0.744	-1.096
2	0.408	3.653	0.664	-2.274
3	0.705	3.142	0.122	-3.892**
4	0.549	3.807	0.182	-4.036*
5	0.086	4.526	-0.305	-3.514*
6	0.279	6.485*	0.183	-3.566
7	0.019	6.127	0.374	-5.208**
8	0.321	4.993	0.062	-5.039*
9	0.243	4.179	-0.282	-6.808**
10	0.218	4.559	-0.577	-6.668**
15	-0.768	-2.342	0.924	-5.927*
20	-0.036	-2.272	-1.066	-5.247

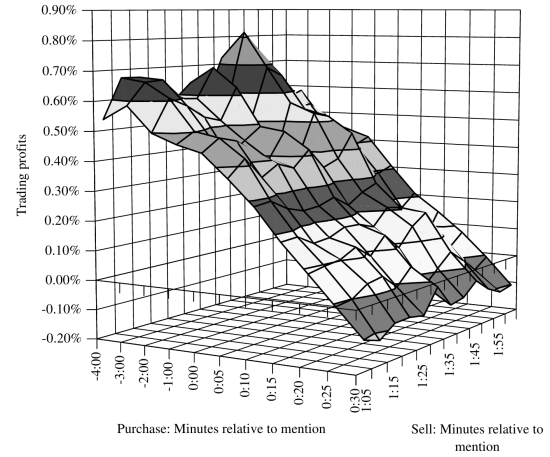


Fig. 2. Trading profits around CNBC reports. The figure shows the percentage price change in average transaction prices surrounding positive on-air stock reports during the Midday Call segment on CNBC. We calculate average purchase prices in intervals from five minutes before the report to 30 seconds after. The interval labeled -4:00 corresponds to the average purchase price from five minutes before the event to four minutes before the event. We calculate average sell prices in intervals from one to two minutes after the event. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

Table 4: Daily cumulative abnormal returns

Figure 2: Trading profits

7.7 Table 5: Trading intensity around CNBC reports

Read:

- The table reports trades per minute and volume per minute around the reports.
- Trading intensity rises substantially after positive reports, especially for *Midday Call* Nasdaq stocks.
- Volume also rises after positive *Midday Call* reports.
- Negative reports show smaller and less uniform trading-intensity changes, partly because the negative sample is small.

Economic message: Price discovery is accompanied by a burst of trading. Information does not passively enter prices; it is incorporated through aggressive orders.

7.8 Table 6: Order imbalance around CNBC reports

Read:

- Positive reports generate positive order imbalances, especially in the first minute.

- Negative reports generate negative order imbalances, especially in the first minute.
- For positive *Midday Call* reports, the first-minute imbalances imply that roughly 70–75% of trades are buyer-initiated.
- The evidence is consistent across NYSE and Nasdaq stocks.

Economic message: The direction of order flow validates the information interpretation. Traders are buying after good news and selling after bad news.

Table 5
Trading intensity around CNBC reports
The table shows the median number of trades and trading volume per minute around on-air stock reports during the Morning Call and Midday Call segments on CNBC. We partition reports into groups depending on whether the report is positive or negative and whether the stock is listed on the NYSE or Nasdaq. One and two stars denote that the median is different from the –10 to –5 minute period median using the one-tailed Kolmogorov-Smirnov test at the five and one percent level, respectively. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

Minutes relative to report	Positive reports				Negative reports	
	Morning Call		Midday Call		NYSE	Nasdaq
	NYSE	Nasdaq	NYSE	Nasdaq		
<i>Panel A: Trades per minute</i>						
–10 to –5	3.8	56.0	2.6	27.2	2.2	20.6
–5 to 0	3.8	62.1	2.8	48.6	3.4	22.6
1	6.0*	58.0	11.0**	286.0**	4.0	37.0*
2	6.0**	46.0	10.0**	183.0**	5.0	42.0
3	6.0**	51.0	8.5**	126.0**	4.0	19.0
4	5.0*	47.5	8.0**	85.0**	3.0	51.0
5	5.0	40.0	6.5**	92.0**	3.0	38.0
10	4.2	52.5	6.0**	52.8*	4.4	43.6
15	4.7	52.9	4.9*	49.2*	2.6	38.2
<i>Panel B: Volume per minute</i>						
–10 to –5	5,850	29,560	3,220	12,960	3,860	7,640
–5 to 0	6,470	30,880	4,940	20,840	2,140	11,640
1	4,650	30,950	19,000**	116,800**	7,400	13,600
2	5,400	28,450	16,150**	74,400**	4,800	17,600
3	5,150	23,350	8,850**	67,700**	4,700	14,100
4	5,250	27,650	7,500**	33,300**	1,600	17,100
5	6,650	25,150	5,800	31,300	2,700	13,700
10	6,710	29,440	6,180**	31,600	6,840	15,180
15	6,100	23,460	6,990**	25,500	3,880	11,480

Table 5: Trading intensity

Table 6
Order imbalance around CNBC reports
The table shows the median order imbalance, defined as (number of buys–number of sells)/(number of buys+number of sells), around on-air stock reports during the Morning Call and Midday Call segments on CNBC. We partition reports into groups depending on whether the report is positive or negative and whether the stock is listed on the NYSE or Nasdaq. One and two stars denote that the median order imbalance is different from the –10 to –5 minute period median using the one-tailed Kolmogorov-Smirnov test at the five and one percent level, respectively. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

Minutes relative to report	Positive reports				Negative reports	
	Morning Call		Midday Call		NYSE	Nasdaq
	NYSE	Nasdaq	NYSE	Nasdaq		
–10 to –5	0.000	–0.100	0.143	0.044	–0.083	–0.097
–5 to 0	0.000	–0.064	0.277*	0.067*	0.292*	–0.204
1	0.200**	0.200**	0.500**	0.398**	–0.487*	–0.330*
2	0.077*	0.023*	0.234	0.009	–0.455**	–0.102
3	0.059*	0.000*	0.143	–0.051	–0.333	–0.292
4	0.200**	–0.111	0.000	0.000	–0.053	–0.325**
5	0.100**	0.077**	0.000	–0.072	–0.067	–0.240
10	0.063	0.011**	0.134	0.058	–0.250	–0.165
15	0.148*	0.032*	0.122	0.042	–0.169	–0.013

Table 6: Order imbalance

7.9 Table 7: Fraction of NYSE-listed trades executed off the NYSE

Read:

- The table focuses on NYSE-listed stocks and measures the fraction of trades executed away from the NYSE floor.
- After positive *Midday Call* reports, the off-NYSE fraction rises significantly in the first several minutes.
- The pattern is consistent with traders using venues that offer faster execution rather than waiting for possible price improvement on the NYSE floor.

Economic message: The trading-feasibility evidence supports the idea that speed is valuable. Traders appear to route orders in ways that prioritize immediate execution when news arrives.

Table 7

Fraction of trades of NYSE-listed stocks executed off the NYSE

The table shows the median fraction of trades of NYSE-listed stocks that execute off the NYSE around on-air stock reports during the Morning Call and Midday Call segments on CNBC. We partition reports into groups depending on whether the report is positive or negative. One and two stars denote that the median percentage is different from the -10 to -5 minute period median using the one-tailed Kolmogorov-Smirnov test at the five and one percent level, respectively. The data cover 322 stock reports from June 12, 2000 through October 27, 2000.

Minutes relative to report	Positive reports		Negative reports
	Morning Call	Midday Call	
-10 to -5	0.364	0.346	0.344
-5 to 0	0.383	0.289	0.330
1	0.414	0.538**	0.452
2	0.500	0.500**	0.500
3	0.500*	0.600**	0.545
4	0.500*	0.500**	0.605*
5	0.500*	0.545**	0.500
10	0.484*	0.450**	0.367
15	0.396	0.387	0.347

Table 7: Off-NYSE execution fraction

8 Short summary

- The paper studies 322 CNBC analyst reports aired during market hours in 2000.
- It combines hand-recorded second-level broadcast timing with TAQ intraday trades and quotes.
- Positive *Midday Call* reports have a large, immediate price impact.
- Positive-news price adjustment is largely complete within one minute.
- Negative reports move prices downward more gradually, over roughly 15 minutes.
- Trading intensity rises sharply after reports, especially after positive *Midday Call* reports.
- Order imbalances move in the direction predicted by the report sentiment.
- Traders who execute within about 15 seconds after positive *Midday Call* reports can earn small but significant profits after trading costs.
- The profitable trading window disappears almost immediately.
- The paper’s main contribution is to measure market efficiency in real time, using seconds rather than days.

9 Conclusion

9.1 Core conclusion

Busse and Green show that stock prices incorporate widely disseminated analyst information very quickly, but not instantaneously. Positive CNBC *Midday Call* reports are mostly reflected in prices within about one minute, while negative reports take longer to be incorporated. The market is therefore highly efficient in real time, but the adjustment process is visible at the second-by-second level.

9.2 Economic intuition

The economic intuition is that financial information becomes valuable only if traders can process and act on it before others do. The CNBC reports create a short public signal. Fast traders buy or sell immediately, their orders create price pressure in the direction of the news, and the price change removes the profit opportunity. Because monitoring CNBC, processing the signal, and routing orders quickly are costly, the small profit available to the fastest traders can be interpreted as compensation for information production and market monitoring.

9.3 Incremental contribution revisited

The paper advances the market-efficiency literature in three ways. First, it moves from daily or multi-minute event windows to second-level evidence. Second, it studies a real-time media channel rather than scheduled corporate announcements or after-hours analyst reports. Third, it connects price adjustment to trading behavior and feasible trading profits. Together, these contributions show not only that prices adjust quickly, but how that adjustment happens through active trading.

9.4 Implications for market efficiency

The paper supports a practical version of semi-strong-form efficiency. Public information is incorporated so quickly that ordinary delayed trading strategies are unlikely to profit. But the existence of a few seconds of profitability also shows why active monitoring and fast execution matter. Markets become efficient because some traders have incentives to make them efficient.

9.5 Limitations

The reports are not randomly selected, and some information may have leaked to analysts' clients before the broadcast. Sentiment classification is partly subjective. Negative reports are relatively few, limiting power for negative-news tests. The sample consists mainly of large, liquid stocks, so the speed estimates may not generalize to small or illiquid firms. Finally, the trading-profit calculations approximate feasible execution but cannot perfectly reproduce every trader's routing technology, latency, and costs.

9.6 Takeaway

The enduring takeaway is that market efficiency is a real-time process. Prices do not wait until the close, the next day, or even the next five minutes. For informative positive news about liquid stocks, the market can largely finish adjustment within one minute, and the trading opportunity for outsiders is measured in seconds.